offering. First, the platform should not be a mere version control tool but must offer rich collaboration features for software development teams. These should include issue tracking, forums, email lists, documentation and releases. Second, the application must be very customisable. It must also have a vibrant community with frequent releases to ensure that NeGD got timely support for any issues. Such support, updates or bug fixes also needed to be well documented if NeGD planned to customise and own them. Last, but not the least, NeGD wanted to select a platform built using tools and technologies that its technical team was conversant with.

Evaluating SourceForge, GitLab and Tuleap

To achieve what was planned, the NeGD team evaluated and experimented with SourceForge, GitLab and Tuleap. But Tuleap emerged as the 'more suitable' in the list. "We evaluated and experimented with SourceForge, GitLab and Tuleap. And, finally, we found Tuleap the most suitable as per our evaluation criteria," Savant says.

Attracting 70 projects in less than 100 days

OpenForge currently hosts over 70 active projects on its platform, and over 60 per cent of the total projects are in the public domain. Some of the important projects by the Ministry of Electronics and IT (MeitY) and NeGD, including DigiLocker and Government eMarket, are already using this platform. Besides, the OpenForge project itself is hosted on the online platform.

Systematic change

Debabrata Nayak, project director of open source collaboration at NeGD, considers OpenForge as a systematic change that required the support of the administration. Nayak, who also leads the technical team of the DigiLocker project, highlights that his team received complete support from the MeitY and NeGD.

Apart from the departmental support, it was the open source policy of the government that helped the development of OpenForge. As part of the Digital India initiative, the Indian government had released the policy titled 'Collaborative Application Development by Opening the Source Code of Government Applications' back in 2015. "The policy provided a solid foundation for OpenForge. It listed the basic principles and elements for building the envisioned platform, and provided a clear idea of what the result ought to be," states Nayak.

Administrative challenges before the launch

While the open source policy enabled the speedy process of building OpenForge, it was difficult to highlight the significance of the initiative in order to create a buy-in from other government departments. "The main challenge was that the idea itself is of a very technical nature and fails to resonate with most of the decision makers," recalls Chauhan.

What OpenForge takes from the government's open source policy

- *Collaboration:* The platform must support a rich set of collaboration features for various roles within a software team.
- *Code repository and version control:* Since the platform will be used by various government organisations to host existing projects, OpenForge must support the most common version control tools like Git and SVN.
- *Support for public and private repositories:* Since the policy anticipated that not every project can be open source, the platform must support public as well as private repositories.
- *Community:* The platform must also support the contributions from the community, academia and industry.

The technical team very often fails to explain the tangible benefits of such an initiative. Therefore, the team used a sample figure by quoting the software applications developed within NeGD using an open source stack. "Once we showed the benefits in terms of the money saved, it was very easy to highlight the importance of the initiative," Chauhan told *Open Source For You*.

Once the launch was planned, the team began the internal testing phase for OpenForge. The prime objective was to "create a simple product without compromising on rich features."

"We wanted OpenForge to be simple for novice users yet provide flexibility to all project teams to achieve what they want," says Amit Ranjan, project architect for OpenForge, NeGD.

How it differs from GitHub

One question raised by the Indian open source community is: What makes OpenForge different from GitHub?

The open source policy by the government encourages the use of open source in as many e-governance applications as possible. But not every e-governance project is going to be open source. Some may remain private. OpenForge has thus been created to provide a collaboration platform for both these public and private projects. The open source projects can very well be posted on GitHub, but enterprise level private repositories are not free on GitHub; they are available at a cost. Moreover, the team wanted the repositories of private projects to be hosted within India. It has therefore leveraged the state-run data centre provided by National Informatics Centre (NIC), which may not have been possible with GitHub. This is the prime purpose of not using GitHub and creating a separate platform.

"Feature wise, GitHub is more a code sharing and version control platform. It keeps on receiving new features continuously. Compared to GitHub, OpenForge supports more rich collaboration features such as various trackers to track bugs and requirements, documentation, forums, mailing lists and release management," Ranjan emphasises.